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CUET UG

Geography

People: Population & Human Development

Fundamentals of Human Geography: Distribution, Density, Growth, Composition, Demographic Transition, and Human Development

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Module 1: World Population Distribution, Density & Growth

1.1 Patterns of World Population Distribution

Human population is distributed with acute spatial unevenness across the globe. As American geographer **George B. Cressey** famously observed: ""Asia has many places where people are few and few places where people are very many.""

This profound remark applies to the entire planet: roughly **90% of the world's population lives on only about 10% of its land area**, while vast expanses of frozen tundra, hyper-arid deserts, and rugged high-altitude mountains remain virtually uninhabited.

The Ten Most Populous Countries: The world's top ten most populous nations together account for roughly **60% of humanity**. Remarkably, **six of these ten nations are located in Asia**:

1. **China** (Asia)
2. **India** (Asia)
3. **United States of America** (North America)
4. **Indonesia** (Asia)
5. **Pakistan** (Asia)
6. **Brazil** (South America)
7. **Nigeria** (Africa)
8. **Bangladesh** (Asia)
9. **Russia** (Eurasia)
10. **Mexico** (North America)

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CUET HIGH-YIELD: DISTRIBUTION BENCHMARKS

- **Asian Dominance:** 6 out of the 10 most populous nations are in Asia.
- **Hemispheric Asymmetry:** Over 85% of global population resides in the Northern Hemisphere.
- **Altitude Concentration:** Over 80% of humankind lives below 500 meters above sea level.

1.2 Density of Population

Population density provides a standardized measure of human concentration relative to physical land space. In NCERT Class 12, the cardinal index is **Arithmetic Population Density**:

ARITHMETIC POPULATION DENSITY FORMULA

$$\text{Density of Population} = \frac{\text{Total Population}}{\text{Total Area}}$$

- Expressed strictly as **persons per square kilometer** ($\frac{\text{persons}}{\text{km}^2}$).
- **Example:** If an area of 100km^2 supports a population of $150,000$ persons:

$$\text{Density} = \frac{150000}{100} = 1,500 \frac{\text{persons}}{\text{km}^2}$$

Global Density Zones:

1. **Densely Populated Zones (Over 200 persons/km^2):**
   - **East and South Asia:** River valleys of the Ganga, Brahmaputra, Yangtze, and Hwang Ho with rich alluvial soils supporting multi-cropping.
   - **North-Western Europe:** Britain, Germany, France, Belgium, and Netherlands—fueled by the Industrial Revolution, coal basins, and urban-commercial agglomeration.
   - **North-Eastern North America:** The Great Lakes megalopolis (Boston to Washington, D.C. - "BosWash") and Saint Lawrence lowlands.
2. **Moderately Populated Zones (11 to 50 persons/km^2):**
   - Western China, Southern India, Norway, Sweden, interior grasslands (Steppes and Prairies).
3. **Sparsely Populated Zones (Under 1 person/km^2):**
   - Extreme biomes: **Hot deserts** (Sahara, Thar, Kalahari, Atacama); **Cold deserts** (Greenland, Arctic tundra, Antarctica); **Equatorial rainforests** (Amazon and Congo basins with leached soils and enervating humidity); and **High mountains** (Himalayas, Andes, Rockies).

1.3 Determinants of Population Distribution

Human locational decisions are shaped by a complex interplay of physical, economic, and socio-cultural forces:

1. **Geographical Factors:**
   - **Availability of Fresh Water:** The most critical requirement for human life, domestic hygiene, livestock, agriculture, and industry. All ancient river-valley civilizations (Nile, Indus, Tigris-Euphrates, Hwang Ho) blossomed along freshwater streams.
   - **Landforms (Relief):** Flat alluvial plains and gentle river terraces favor road construction, canal networks, and mechanized farming. Conversely, steep mountain slopes impede transport, increase soil erosion, and hinder urban settlements.
   - **Climate:** Extreme climates (hyper-arid heat or sub-zero arctic blizzards) discourage human habitation. Regions with comfortable, sub-tropical or **Mediterranean climates** attract dense human settlements.
   - **Soils:** Fertile loamy and alluvial soils support intensive intensive agriculture, sustaining high population densities.
2. **Economic Factors:**
   - **Minerals:** Mining and smelting operations attract capital, engineers, and laborers. A classic NCERT benchmark is the **Katanga-Zambia copper belt** in central Africa.
   - **Urbanization:** Cities offer diversified employment, superior healthcare, advanced universities, and civic amenities. Rural-to-urban distress migration fuels massive agglomerations.
   - **Industrialization:** Manufacturing belts provide vast employment opportunities not only in factory floors but also in transport, commerce, and banking. The **Kobe-Osaka industrial region of Japan** is a classic NCERT example.

3. **Social and Cultural Factors:**
   - Sacred religious cities (Varanasi, Jerusalem, Mecca, Rome) attract pilgrims and permanent settlers.
   - Conversely, political instability, civil wars, and ethnic cleansing trigger massive refugee outflows.

Factor Category	Primary Determinants	NCERT High-Yield Benchmark Examples
Geographical	Fresh water, flat relief, moderate climate, alluvium.	Indo-Gangetic plain; Mediterranean coastal belt.
Economic: Minerals	Mineral deposits, mining infrastructure.	Katanga-Zambia copper belt (Central Africa).
Economic: Industry	Factory clusters, tertiary agglomeration.	Kobe-Osaka industrial region (Japan).
Social & Cultural	Religious sanctity, cultural heritage, peace.	Varanasi, Jerusalem, Mecca; flight from war zones.

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Module 2: Components of Population Change & Spatial Dynamics

2.1 The Three Components of Population Change

Population change represents the dynamic shift in the number of inhabitants in a territory over a defined timeframe. It is governed strictly by three variables: **Births, Deaths, and Migration**.

VITAL RATES FORMULAS

- **Crude Birth Rate (CBR):**

$$\text{CBR} = \frac{\text{Bi}}{\text{P}} \times 1000$$

  Where Bi = Number of live births during the year; P = Estimated mid-year population.
- **Crude Death Rate (CDR):**

$$\text{CDR} = \frac{\text{D}}{\text{P}} \times 1000$$

  Where D = Number of deaths during the year; P = Estimated mid-year population.
- **Natural Growth of Population:**

$$\text{Natural Growth} = \text{Births} - \text{Deaths}$$

- **Actual Growth of Population:**

$$\text{Actual Growth} = (\text{Births} - \text{Deaths}) + (\text{In-migration} - \text{Out-migration})$$

2.2 Migration: Spatial Movement & Selectivity

Migration is the physical movement of people from a **Place of Origin** to a **Place of Destination**. It can be permanent, semi-permanent, or seasonal.

- **Immigration:** Migrants who move into a new place (In-migrants).
- **Emigration:** Migrants who leave their home territory (Out-migrants).

The Two Sets of Migration Drivers: CUET UG regularly frames distinction questions around **Push Factors** and **Pull Factors**:

Dimension	Push Factors (Place of Origin)	Pull Factors (Place of Destination)
Core Definition	Negative pressures making home origin unbearable.	Positive attractions drawing migrants to a new place.
Economic Drivers	Severe unemployment, agrarian distress, poverty.	Lucrative job opportunities, higher real wages.
Environmental Drivers	Droughts, floods, soil exhaustion, earthquakes.	Pleasant climate, abundant water, fertile terrain.
Social & Political	Ethnic riots, civil war, political tyranny, casteism.	Peace, rule of law, security of life and property.
Civic Infrastructure	Poor schools, absent healthcare, primitive housing.	Modern hospitals, prestigious universities, entertainment.

2.3 Trends in Population Growth & The Malthusian Theory

The human population grew at an extraordinarily sluggish rate for millennia. It took thousands of years to reach its first **1 billion in the year 1830**. However, the advent of the Industrial Revolution, modern medicine, public sanitation, and mechanical agriculture ignited an unprecedented acceleration:

- **1830:** 1 billion
- **1930:** 2 billion (Doubling time: 100 years)
- **1960:** 3 billion (Added 1 billion in 30 years)
- **1975:** 4 billion (Added 1 billion in 15 years)
- **1987:** 5 billion (Added 1 billion in 12 years)
- **1999:** 6 billion (Added 1 billion in only 12 years!)

THOMAS MALTHUS THEORY OF POPULATION (1798)

English political economist **Thomas Robert Malthus**, in his landmark work **An Essay on the Principle of Population** (1798), warned:

- **Divergence:** Human population grows **geometrically** $(1, 2, 4, 8, 16, 32...)$, whereas agricultural food production expands only **arithmetically** $(1, 2, 3, 4, 5, 6...)$.
- **Malthusian Catastrophe:** If humanity does not exercise **preventive checks** (delayed marriage, moral restraint, birth spacing), nature will inevitably impose ruthless **positive checks** (famines, pestilence, epidemic diseases, and war) to restore ecological balance.

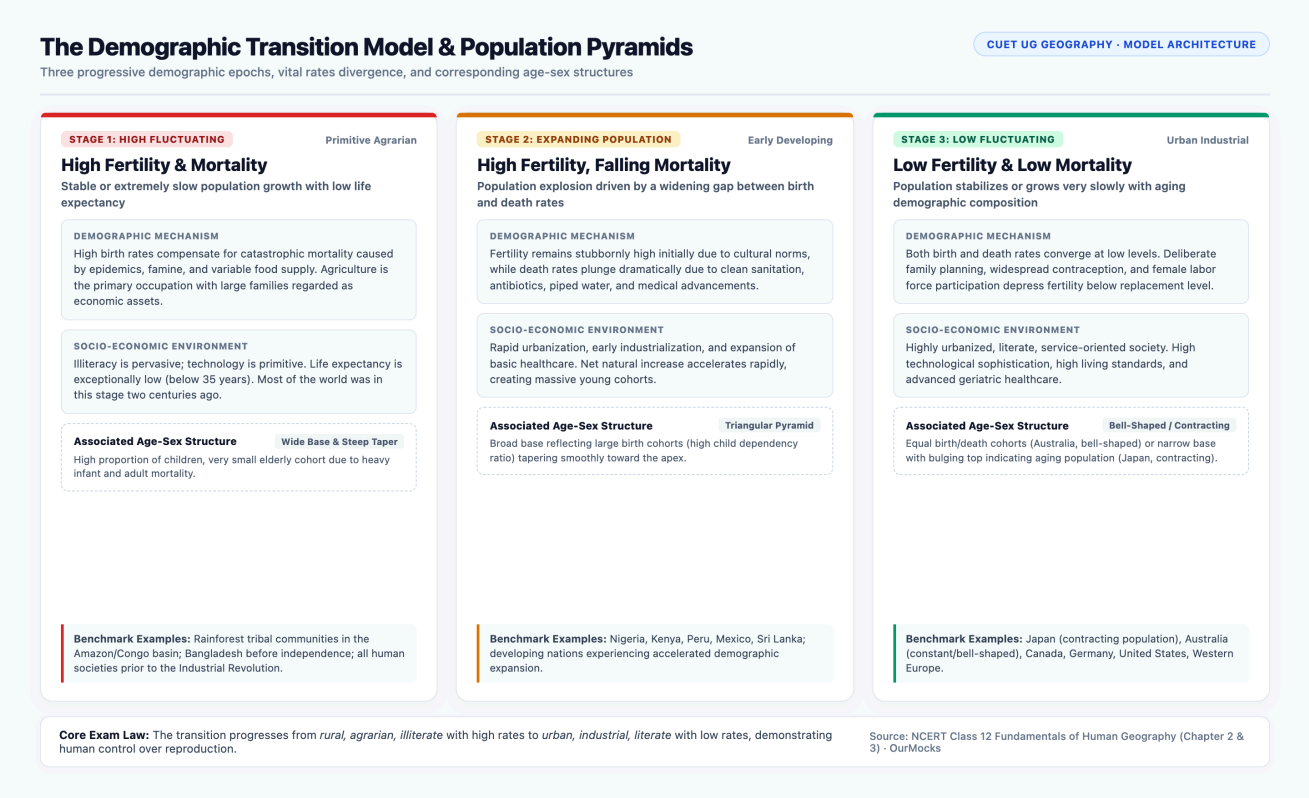


Figure 1: The Three Stages of the Demographic Transition Model and Associated Age-Sex Pyramids

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Module 3: Demographic Transition Theory & Population Composition

3.1 The Demographic Transition Model (DTM)

Demographic Transition Theory describes and predicts the progressive demographic trajectory of any human society as it modernizes. The core axiom of DTM states that: ""Human societies transition progressively from a high-birth, high-death, rural, agrarian, and illiterate state to a low-birth, low-death, urban, industrial, and literate state.""

The transition unfolds through three distinct stages:

1. **Stage 1: High Fluctuating (Primitive Agrarian Society)**
   - **Vital Rates:** Both birth rate and death rate are exceptionally high (> 35 per 1,000). Population growth is sluggish or stagnant.

- **Ecological Context:** High mortality caused by epidemics, variable harvests, and malnutrition. High fertility compensates for heavy infant deaths. Life expectancy is below 35 years.
- **Benchmark:** All societies 200 years ago; remote rainforest tribal groups today.

2. **Stage 2: Expanding Population (Early Developing Society)**
   - **Vital Rates:** Fertility remains high initially, but mortality plunges precipitously.
   - **The Expansion Gap:** Introduction of clean water, vaccination, antibiotics, and sewage management causes death rates to collapse. The resulting wide gap between high CBR and falling CDR causes a massive **population explosion**.
   - **Benchmark:** Developing nations (Nigeria, Kenya, Bangladesh, Peru, Sri Lanka).
3. **Stage 3: Low Fluctuating / Contracting (Advanced Urban Society)**
   - **Vital Rates:** Both birth and death rates converge at very low levels (< 12 per 1,000). Population stabilizes or grows extremely slowly.
   - **Ecological Context:** Highly urbanized, literate society. Universal family planning, high female workforce participation, and high cost of child-rearing depress fertility below replacement level (2.1). In some nations, birth rates dip below death rates, creating population decline.
   - **Benchmark:** Japan (contracting), Australia (constant/bell-shaped), Germany, Canada, USA.

3.2 Age-Sex Pyramids (Population Pyramids)

An **Age-Sex Pyramid** graphically portrays the demographic composition of a country. The percentage of males is plotted on the left, and females on the right, broken into five-year age brackets:

Pyramid Type	Geometric Profile	Demographic Characteristics	NCERT Benchmark Country
Expanding Population	Triangular shaped; broad base with tapered top.	High birth rates; large young cohort; high child dependency ratio.	**Nigeria**, Bangladesh, Mexico
Constant Population	Bell-shaped; symmetrical; sides taper near apex.	Birth and death rates nearly equal; stationary population.	**Australia**
Declining Population	Narrow base with tapered top and bulging middle.	Low birth and death rates; aging population; labor shortages.	**Japan**

3.3 Sex Ratio & Global Disparities

Sex ratio is the index of numerical balance between men and women.

- **Calculation in India:** Number of females per 1,000 males:

$$\text{Sex Ratio} = \frac{\text{Number of Females}}{\text{Number of Males}} \times 1000$$

- **Calculation in World / UN Practice:** Number of males per 100 or 1,000 females:

$$\text{Sex Ratio (World)} = \frac{\text{Number of Males}}{\text{Number of Females}} \times 1000$$

CUET HIGH-YIELD: GLOBAL SEX RATIO ANOMALIES

- **World Average:** Roughly 102 males per 100 females.
- **Highest Female Ratio: Latvia** (85 males per 100 females / 1,187 females per 1,000 males), caused by heavy male mortality in World War II and high male adult cardiovascular mortality.
- **Lowest Female Ratio: United Arab Emirates (UAE)** and Qatar (300+ males per 100 females), caused by massive selective immigration of male foreign contract laborers.

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Module 4: The Human Development Paradigm (Haq & Sen)

4.1 Growth vs. Development: The Paradigmatic Shift

For decades, a country's development was judged purely by economic growth—its Gross National Product (GNP) and per capita income. In 1990, Pakistani economist **Dr. Mahbub-ul-Haq** and Indian Nobel Laureate **Prof. Amartya Sen** dismantled this narrow view:

Dimension	Economic Growth	Human Development
Nature of Change	Purely **quantitative** change; monetary expansion.	**Qualitative** change; positive transformation in life.
Value Neutrality	**Value-neutral**; growth can be positive or negative.	**Value-positive**; occurs only when quality of life improves.
Primary Focus	Production of commodities, infrastructure, and GDP.	Expanding human capabilities, freedoms, and choices.
Central Question	"How much is the nation producing?"	"Are citizens healthy, educated, and free to choose?"

""Human development is development that enlarges people's choices and improves their lives. People are the real wealth of nations. The basic objective of development is to create an environment where people can enjoy long, healthy, and creative lives.""

4.2 The Four Pillars of Human Development

Just as a palace rests on structural pillars, human development rests on four foundational concepts:

1. **Equity:** Ensuring equal access to opportunities available to everybody, regardless of gender, race, income, caste, or religion.

2. **Sustainability:** Continuity in the availability of opportunities for future generations. All environmental, financial, and human resources must be utilized with regenerative stewardship.
3. **Productivity:** Building human labor productivity through continuous investment in education, technical skilling, and healthcare.
4. **Empowerment:** Having the freedom and capability to make meaningful choices. Requires good governance, democratic decentralization, and affirmative support for marginalized groups.

4.3 The Four Approaches to Human Development

Human development methodologies evolved through four historical approaches:

1. **Income Approach:** The oldest approach; links development directly to personal income. Higher income reflects higher freedom.
2. **Welfare Approach:** Views human beings as passive targets of government largesse; argues for maximizing public spending on health, education, and amenities.
3. **Basic Needs Approach:** Formulated by the **International Labour Organisation (ILO)**. Identifies six essential basic needs: **Health, Education, Food, Water Supply, Sanitation, and Housing**.
4. **Capability Approach:** Pioneered by **Prof. Amartya Sen**. Focuses on building human capabilities in education, health, and access to resources to unlock true human agency and freedom.

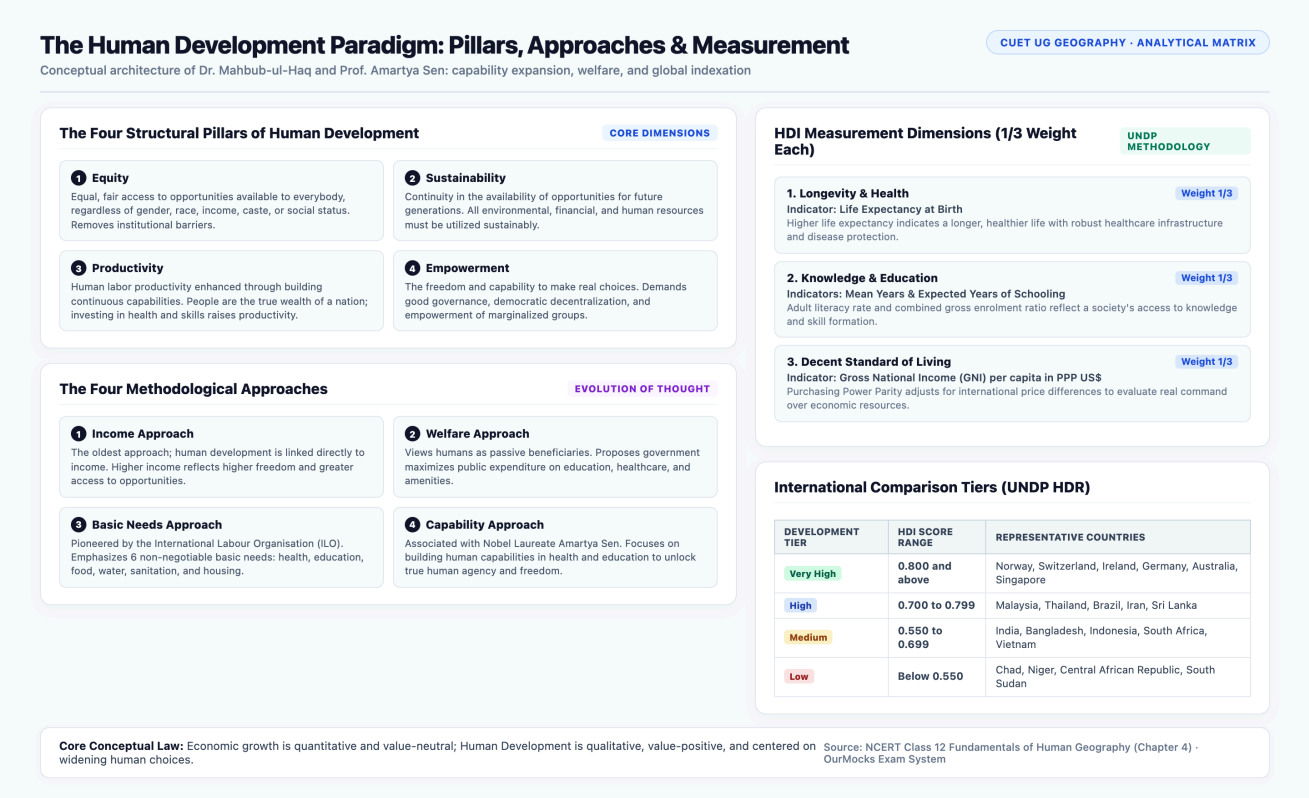


Figure 2: The Human Development Matrix: The Four Pillars, Four Approaches, and HDI Dimensions

Module 5: Measuring Human Development & Global Indexation

5.1 Measuring the Human Development Index (HDI)

Published annually since 1990 by the United Nations Development Programme (UNDP), the **Human Development Index (HDI)** measures average achievement in three basic dimensions of human development:

Dimension (Weight: 1/3)	Key Statistical Indicator	Significance
1. Long & Healthy Life	**Life Expectancy at Birth**	Evaluates survival chances and the strength of healthcare infrastructure.
2. Access to Knowledge	**Mean & Expected Years of Schooling**	Evaluates adult literacy and combined educational enrolment ratios.
3. Decent Living Standard	**Gross National Income (GNI) per capita in PPP USD**	Adjusts for international price parity to measure real purchasing power.

HDI SCORING AND WEIGHTAGE

- Each of the three dimensions is assigned an equal weight of $\frac{1}{3}$.
- The composite score ranges from **0.000 to 1.000**. The closer the score is to 1.0, the higher the level of human development.
- **Human Poverty Index (HPI):** Measures non-income shortfalls in human development (probability of not surviving to age 40, adult illiteracy rate, unweighted average of lack of access to clean water and underweight children).

5.2 International Comparisons & Development Tiers

UNDP classifies countries into four distinct human development tiers based on composite scores:

Development Tier	Score Range	Representative Countries
Very High	0.800 and above	Norway, Switzerland, Ireland, Germany, Australia, Singapore
High	0.700 to 0.799	Malaysia, Thailand, Brazil, Iran, Sri Lanka
Medium	0.550 to 0.699	India, Bangladesh, Indonesia, South Africa, Vietnam
Low	Below 0.550	Niger, Chad, Central African Republic, South Sudan

THE SIZE-PARADOX IN HUMAN DEVELOPMENT

A country's HDI rank is **not determined by the physical size of its territory or the magnitude of its GDP**:

- **Sri Lanka** and **Maldives** frequently rank higher than India, despite having vastly smaller economies.
- Within India, **Kerala** achieves HDI levels comparable to advanced nations due to historical, state-sponsored investments in universal primary education and primary healthcare.
- Nations that spend heavily on social sectors rather than military armaments consistently attain higher human development.

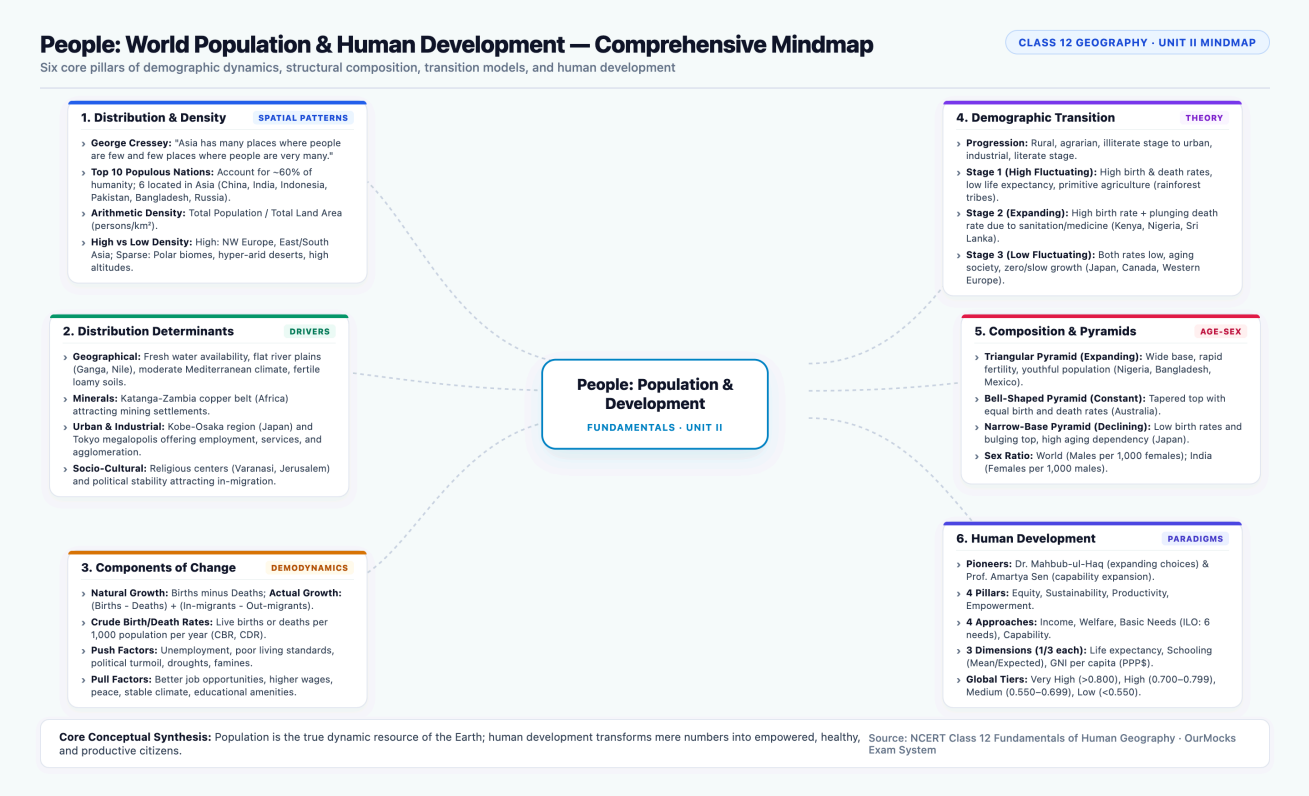


Figure 3: Comprehensive Summary Mindmap: World Population Dynamics, DTM, and Human Development

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Module 6: High-Yield CUET Comparison Tables & Concept Synthesis

6.1 The 4 Pillars vs. The 4 Approaches Cross-Matrix

Analytical Framework	Component	Core Disciplinary Focus
The 4 Pillars	Equity	Equal access to opportunities regardless of gender, race, or caste.

Analytical Framework	Component	Core Disciplinary Focus
The 4 Pillars	Sustainability	Continuity in resource availability for future generations.
The 4 Pillars	Productivity	Enhancing human labor capability through health and skilling.
The 4 Pillars	Empowerment	Expanding freedom to make choices through good governance.
The 4 Approaches	Income Approach	Oldest model; links human development directly to income levels.
The 4 Approaches	Welfare Approach	Government maximizes social spending; humans are passive targets.
The 4 Approaches	Basic Needs (ILO)	Guarantees 6 essentials: health, education, food, water, sanitation, shelter.
The 4 Approaches	Capability (Sen)	Building human capabilities unlocks true freedom and agency.

6.2 Key Formulas and Demographic Metrics Cheat-Sheet

Demographic Metric	Standard Formula	Unit of Measure
Arithmetic Density	$\frac{\text{Total Population}}{\text{Total Area}}$	$\frac{\text{Persons}}{\text{km}^2}$
Crude Birth Rate (CBR)	$\left(\frac{\text{Bi}}{P}\right) \times 1000$	Per 1,000 population
Crude Death Rate (CDR)	$\left(\frac{D}{P}\right) \times 1000$	Per 1,000 population
Natural Growth Rate	Births − Deaths	Net natural balance
Actual Growth Rate	$(B - D) + (\text{In-migrants} - \text{Out-migrants})$	Total net balance
Sex Ratio (India)	$\left(\frac{\text{Number of Females}}{\text{Number of Males}}\right) \times 1000$	Females per 1,000 males
Sex Ratio (UN / World)	$\left(\frac{\text{Number of Males}}{\text{Number of Females}}\right) \times 1000$	Males per 1,000 females

Module 7: Master Revision Cheatsheet & Demographic Metrics

MASTER REVISION MATRIX — POPULATION DYNAMICS & HUMAN DEVELOPMENT

Concept / Metric	Exact Formula / Definition	Key Demographic Significance
Arithmetic Density	$\frac{\text{Total Population}}{\text{Total Land Area}}$	Physical man-land pressure ($\frac{\text{persons}}{\text{km}^2}$). World avg $\approx$ 54; Monaco, Bangladesh top.
Crude Birth Rate (CBR)	$\left(\frac{\text{Bi}}{P}\right) \times 1000$	Number of live births per 1,000 mid-year population in an area.
Crude Death Rate (CDR)	$\left(\frac{D}{P}\right) \times 1000$	Number of deaths per 1,000 mid-year population in an area.
Natural Growth Rate	Births − Deaths	Excludes international and inter-regional migration flows.
Actual Growth Rate	$(B - D) +$ (In-migration − Out-migration)	Complete demographic balance equation incorporating net spatial mobility.
Sex Ratio (India)	$\left(\frac{\text{Number of Females}}{\text{Number of Males}}\right) \times 1000$	Females per 1,000 males. Unfavorable in societies with female foeticide/neglect.
Sex Ratio (UN / World)	$\left(\frac{\text{Number of Males}}{\text{Number of Females}}\right) \times 1000$	Males per 1,000 females (or per 100 females in standard demographic publications).
Demographic Transition	Stage 1 (High CBR/CDR) → Stage 2 (High CBR, Falling CDR) → Stage 3 (Low CBR/CDR)	Transition from primitive agrarian autarky to literate, urban industrial society.
The 4 Pillars of HD	Equity, Sustainability, Productivity, Empowerment	Conceived by Dr. Mahbub ul Haq & Amartya Sen (UNDP 1990).
The 4 Approaches to HD	Income, Welfare, Basic Needs (ILO), Capability (Sen)	Evolution of human development methodology from pure money to capabilities.
HDI Dimensions (1/3 each)	Long & Healthy Life (Life exp 85/20); Knowledge (Mean/Expected schooling); Decent Standard (GNI per capita PPP \$\$)	Geometric mean of the 3 normalized dimensional sub-indices (score 0 to 1).
Gross National Happiness	Bhutan is the only country in the world officially measuring GNH	Prioritizes spiritual, environmental, and cultural well-being over GDP.

THE DEMOGRAPHIC BALANCE EQUATION & DOUBLING TIME LAW

- **Demographic Doubling Time:** At a 1% annual growth rate, a population doubles in approximately 70 years (Doubling Time $\approx \frac{70}{r}$). Rapidly developing nations growing at 2% double every 35 years!
- **Malthusian Natural Checks:** Thomas Malthus (1798) posited that population grows geometrically $(1, 2, 4, 8)$ while food supply grows arithmetically $(1, 2, 3, 4)$, resulting in catastrophic famine, disease, and war (**positive checks**) unless controlled by moral restraint.

Module 8: The Seven Fatal CUET Traps & Diagnostic Rules

THE SEVEN FATAL CUET EXAM TRAPS IN POPULATION & HUMAN DEVELOPMENT

The Trap Concept	The Common Student Slip	The Exact NCERT / NTA Rule
1. Sex Ratio Formula Inversion	Applying the Indian formula when a question asks for the "UN World standard".	India = **Females / Males × 1000**. UN / World practice = **Males / Females × 1000** (or per 100 females)!
2. Natural vs Actual Growth	Including migration flows in calculating "Natural Growth".	Natural Growth is **strictly Births minus Deaths**. Actual Growth **adds net migration** $((B - D) + (I - E))$.
3. Push vs Pull Factors	Classifying "better healthcare/jobs" as Push factors because people are "pushed" to leave.	Push factors = **negative local distress** (war, poverty, droughts). Pull factors = **attractive destination perks** (jobs, security).
4. Four Pillars vs Four Approaches	Treating Pillars and Approaches as interchangeable lists.	Pillars = **Equity, Sustainability, Productivity, Empowerment**. Approaches = **Income, Welfare, Basic Needs, Capability**.
5. HDI vs HPI Orientation	Assuming Human Poverty Index (HPI) measures monetary income poverty.	HPI is a **non-income measure** evaluating shortfalls in health, adult illiteracy, lack of clean water, and underweight children!
6. Country Size vs HDI Rank	Assuming large, wealthy global economies always rank highest in HDI.	NCERT explicitly notes the **Size-Paradox**: small nations (Sri Lanka, Kerala in India) often outrank giant economic neighbors!
7. Malthusian Timeline & Premise	Stating Thomas Malthus wrote in 1898 or praised modern birth control.	Malthus wrote his **Essay on the Principle of Population in 1798**, predicting food arithmetic growth $(1, 2, 3)$ vs population geometric growth $(1, 2, 4, 8)$.

HIGH-FREQUENCY CUET UG TESTING PATTERNS IN DEMOGRAPHY

- **Age-Sex Pyramids Classification:** Expanding = triangular with broad base (Nigeria, Bangladesh); Constant = bell-shaped with tapered top (Australia); Declining = narrow base and tapered top showing low birth/death rates (Japan).
- **HDI Score Ranges:** Very High (≥ 0.800); High ($0.700 - 0.799$); Medium ($0.550 - 0.699$); Low (< 0.550). India consistently falls in the Medium Human Development category.

Module 9: CUET UG Practice Questions — Part 1 (Q1 to Q5)

Q1. According to American geographer George B. Cressey's famous observation on Asian demography:

(A) Asia has fertile plains everywhere supporting identical population concentrations.
(B) Asia has many places where people are few and few places where people are very many.
(C) Asia's population growth is governed purely by industrial development.
(D) Asia has zero regional disparities in arithmetic population density.

Sol. Q1. George B. Cressey famously observed that "Asia has many places where people are few and few places where people are very many," highlighting extreme spatial clustering in fertile river plains alongside vast uninhabited highlands and deserts.
Answer: (B)

Q2. Roughly what percentage of the total world population is concentrated in the top ten most populous countries?

(A) 25%
(B) 40%
(C) 60%
(D) 85%

Sol. Q2. The world's ten most populous countries account for approximately 60% of the world's population. Six of these ten countries (China, India, Indonesia, Pakistan, Bangladesh, Russia) are in Asia.
Answer: (C)

Q3. If a region covers an area of 500km^2 and has a total population of $250,000$ persons, what is its Arithmetic Population Density?

(A) 200 persons/km^2
(B) 500 persons/km^2
(C) 1,000 persons/km^2
(D) 50 persons/km^2

Sol. Q3. Arithmetic Population Density $= \frac{\text{Total Population}}{\text{Total Area}} = 250, \frac{000}{500} = 500 \frac{\text{persons}}{\text{km}^2}$.
Answer: (B)

Q4. Which of the following is cited in NCERT as a classic example of an industrial region that attracts dense population concentrations due to manufacturing and tertiary employment?

(A) Katanga-Zambia copper belt
(B) Kobe-Osaka region of Japan
(C) Atacama Desert of Chile
(D) Amazon basin of South America

Sol. Q4. The Kobe-Osaka region of Japan is a densely populated industrial cluster that attracted millions due to manufacturing, shipping, trade, and commercial employment opportunities.
Answer: (B)

Q5. Which of the following migration drivers is classified strictly as a **Push Factor** operating at the place of origin?

(A) Pleasant sub-tropical climate
(B) High real wages and lucrative career options
(C) Agrarian distress and severe unemployment
(D) Access to modern medical amenities

Sol. Q5. Push factors are negative circumstances that compel people to leave their native place of origin. Severe unemployment, agrarian distress, drought, and lack of basic infrastructure are push factors.
Answer: (C)

Practice Questions — Part 2 (Q6 to Q10)

Q6. In his 1798 work **An Essay on the Principle of Population**, Thomas Robert Malthus stated that:

(A) Population grows arithmetically while food production grows geometrically.
(B) Population grows geometrically while food production grows arithmetically.
(C) Both population and food supplies expand in identical geometric ratios.
(D) Technology will indefinitely expand agricultural yields without ecological checks.

Sol. Q6. Malthus asserted that population increases geometrically $(1, 2, 4, 8, 16...)$ while food production increases only arithmetically $(1, 2, 3, 4, 5...)$, leading to inevitable natural checks like famines and disease.
Answer: (B)

Q7. During which stage of the Demographic Transition Model does a society experience a **population explosion** due to high birth rates coupled with a steep collapse in death rates?

(A) Stage 1: High Fluctuating
(B) Stage 2: Expanding Population
(C) Stage 3: Low Fluctuating
(D) Stage 4: Post-Industrial Contracting

Sol. Q7. In Stage 2 (Expanding), improvements in medical sanitation, clean drinking water, and disease control cause death rates to drop rapidly, while birth rates remain stubbornly high, creating a wide expansion gap and rapid population growth.
Answer: (B)

Q8. A population pyramid with a broad base tapering smoothly to a narrow apex (triangular shape) indicates:

(A) Declining birth rates and an aging population
(B) Equal birth and death rates leading to zero growth
(C) High birth rates and an expanding, youthful population
(D) Severe female deficit due to labor immigration

Sol. Q8. A triangular pyramid with a wide base reflects high birth rates and a large child cohort typical of developing countries like Nigeria, Bangladesh, and Mexico.
Answer: (C)

Q9. Which country's age-sex pyramid is bell-shaped and symmetrical, reflecting nearly equal birth and death rates?

(A) Nigeria
(B) Japan
(C) Australia
(D) Kenya

Sol. Q9. Australia's age-sex pyramid is bell-shaped and symmetrical, tapering slowly toward the top, indicating equal birth and death rates and a stationary population.
Answer: (C)

Q10. The country with the highest proportion of females relative to males in the world (lowest male-to-female ratio) is:

(A) Qatar
(B) United Arab Emirates
(C) Latvia
(D) India

Sol. Q10. Latvia has the world's highest female ratio (roughly 85 males per 100 females / 1,187 females per 1,000 males), attributed to wartime male mortality and higher adult male mortality from health factors.
Answer: (C)

Practice Questions — Part 3 (Q11 to Q15)

Q11. The concept of Human Development was introduced in 1990 by which economist, who defined it as development that enlarges people's choices and improves their lives?

(A) Prof. Amartya Sen
(B) Dr. Mahbub-ul-Haq
(C) Thomas Robert Malthus
(D) David M. Smith

Sol. Q11. Pakistani economist Dr. Mahbub-ul-Haq created the Human Development Index and authored the inaugural 1990 UNDP Human Development Report, defining it as the process of widening human choices and freedoms.
Answer: (B)

Q12. Which of the following is NOT one of the **Four Pillars of Human Development**?

(A) Equity
(B) Sustainability
(C) Industrialization
(D) Empowerment

Sol. Q12. The Four Pillars of Human Development are Equity, Sustainability, Productivity, and Empowerment. Industrialization is an economic process, not a foundational pillar of human development.
Answer: (C)

Q13. The **Basic Needs Approach** to Human Development, which identified six non-negotiable basic needs (Health, Education, Food, Water, Sanitation, Housing), was pioneered by:

(A) International Monetary Fund (IMF)
(B) World Bank
(C) International Labour Organisation (ILO)
(D) United Nations Development Programme (UNDP)

Sol. Q13. The Basic Needs Approach was formulated by the International Labour Organisation (ILO), identifying six vital needs without emphasizing human choice directly.
Answer: (C)

Q14. What equal weight is assigned to each of the three dimensions (Health, Education, Decent Living Standard) in calculating the composite Human Development Index (HDI)?

(A) 1/2
(B) 1/3
(C) 1/4
(D) 1/10

Sol. Q14. In the UNDP HDI methodology, each of the three dimensions (Longevity/Health, Knowledge/Education, and Standard of Living/Income) is assigned an equal weight of 1/3 in calculating the composite geometric score.
Answer: (B)

Q15. According to UNDP classification benchmarks, a country with an HDI composite score of **0.825** is categorized under which development tier?

(A) Low Human Development
(B) Medium Human Development
(C) High Human Development
(D) Very High Human Development

Sol. Q15. Countries with composite HDI scores of 0.800 and above are categorized under the "Very High Human Development" tier. A score of 0.825 falls squarely into this elite tier.
Answer: (D)